

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2077

Badoura 115 138 kV
July 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2077. The Interconnection Customer proposes a 79.7 MW battery energy storage facility interconnecting at the Badoura 115 138 kV substation in the Pennsylvania area. The request was received on May 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 669 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$667,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FILBET – MCREE 138kV	MOUNT LAUREL – MATHEWSON 345kV	103.1	231	6.4
SERRAMONTE – COXS CORNER 138kV	MOUNT LAUREL – MATHEWSON SUBSTATION 345kV Line	102.5	230	7.4
CO LINE RD – BRUNNER ISLAND 500kV	FAYETTE – IRONWOOD LLC 345kV	102.4	2,335	11.5
HILLS CHPEL – BURNS 345kV	CLEAR SPRINGS 500/345kV Transformer	100.3	1,962	7.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MCREE – FILBERT 138kV	Reconductor line	\$2,415,000	5% (\$112,000)
SERRAMONTE – COXS CORNER 138kV	Reconductor line	\$3,976,000	7% (\$268,000)
CO LINE RD – BRUNNER ISLAND 500kV	Reconductor line	\$3,662,000	1% (\$21,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HILLS CHPEL – BURNS 345kV	Reconductor line	\$33,273,000	1% (\$266,000)
	Total allocated to project		\$667,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.